

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Revision Test 2026

PHYSICS — SSC-I (9th Grade)

Syllabus: Unit 1 – Physical Quantities and Measurement, Unit 2 – Kinematics, Unit 3 – Dynamics-I, Unit 4 – Dynamics-II, Unit 5 – Pressure and Deformation, Unit 6 – Work and Energy (to Example 6.4), Unit 9 – Nature of Science and Physics

Total Marks: 38

Time Allowed: 1 Hour 45 Minutes

Version 16 (B)

General Instructions:

1. Attempt **ALL** questions. **There are no choices in this paper.**
2. Section A carries 10 marks, Section B carries 12 marks and Section C carries 16 marks.
3. In Section A, write the **letter** of the correct option clearly. Do the rough working for each MCQ on your answer sheet, then **check that the letter you write matches your own working.** Cutting or overwriting an answer will not be credited.
4. Number your MCQ answers exactly as they are numbered on this paper, and check that all ten are answered.
5. In numericals, show the formula, then the substitution, then the answer **with its unit.** Marks are given for the working.
6. Answer **every part** of each question. Where a question says *state, name, show that* or *explain*, write that sentence out.
7. A calculator **may be used.** Even so, write the formula and the substitution for every numerical, because the marks are for the working. Take $g = 10 \text{ m/s}^2$ unless a question says otherwise.

SECTION A — Multiple Choice Questions ($10 \times 1 = 10$ Marks)

Attempt **ALL** questions. Write the letter of the correct option.

#	Question	A	B	C	D	Answer
1	A cyclist speeds up uniformly from 5 m/s to 20 m/s in 5 s. His acceleration is:	3 m/s ²	5 m/s ²	15 m/s ²	75 m/s ²	(A) (B) (C) (D)
2	The momentum of a 4 kg body moving at 7 m/s is:	11 kg m/s	28 kg m/s	1.75 kg m/s	28 N	(A) (B) (C) (D)
3	According to Newton's third law, the action and reaction forces are equal and opposite. They act:	on the same body	only when the bodies are touching	only if both bodies are at rest	on two different bodies	(A) (B) (C) (D)
4	The single point at which the whole weight of an object appears to act is called its:	centre of motion	pivot	centre of gravity	line of action	(A) (B) (C) (D)
5	The second condition of equilibrium states that the sum of all torques acting on a body must be:	zero	maximum	equal to its weight	equal to the applied force	(A) (B) (C) (D)
6	The pressure exerted by water at a depth of 4 m is (density of water = 1000 kg/m ³ , $g = 10 \text{ m/s}^2$):	400 Pa	4000 Pa	40,000 Pa	400,000 Pa	(A) (B) (C) (D)
7	A force of 20 N acts at a perpendicular distance of 0.3 m from a pivot. The torque produced is:	0.015 N m	6 N m	20.3 N m	66.7 N m	(A) (B) (C) (D)
8	A machine does 1800 J of work in 6 s. Its power is:	10,800 W	1794 W	0.003 W	300 W	(A) (B) (C) (D)
9	A wind turbine generates electricity by converting the:	kinetic energy of moving air into electrical energy	heat energy of the air into electrical energy	chemical energy of the air into electrical energy	potential energy of the air into electrical energy	(A) (B) (C) (D)
10	The study of the physics of respiration, and of radiation used in diagnosis and treatment, belongs to the interdisciplinary	astronomy	biophysics	geophysics	optics	(A) (B) (C) (D)

	field of:					
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SECTION B — Short Questions (3 × 4 = 12 Marks)

Attempt **ALL** questions. There are no choices.

Part	Question	Marks
2(i)	Two forces of 9 N and 12 N act on a body at right angles to each other. (a) Find the magnitude of their resultant. [2] (b) Find the direction of the resultant, measured from the 9 N force. [2]	4
2(ii)	A uniform metre rule is pivoted at its centre. A weight of 6 N hangs 20 cm to the left of the pivot, and an unknown weight W hangs 15 cm to the right. The rule balances. (a) Calculate W. [3] (b) Name the condition of equilibrium you have used. [1]	4
2(iii)	A girl of mass 50 kg climbs a staircase of vertical height 4 m in 8 s . Take $g = 10 \text{ m/s}^2$. (a) Calculate the work she does against gravity. [2] (b) Calculate her power. [2]	4

SECTION C — Long Questions ($2 \times 8 = 16$ Marks)Attempt **ALL** questions. There are no choices.

Part	Question	Marks
3(i)	A car of mass 600 kg starts from rest and reaches a speed of 20 m/s in 10 s , moving in a straight line. (a) Calculate its acceleration. [2] (b) Calculate the distance it travels in these 10 s. [2] (c) Calculate the force acting on the car. [2] (d) Calculate the change in its momentum, and state the unit. [2]	8
3(ii)	In a hydraulic jack, a force of 100 N is applied to a small piston of area 0.004 m² . The large piston has an area of 0.2 m² . (a) Calculate the pressure produced on the small piston. [2] (b) Calculate the force lifted by the large piston. [2] (c) Show that the force has been multiplied by the same number of times as the ratio of the two areas. [2] (d) Explain why a liquid, and not a gas, is used in a hydraulic jack. [2]	8

Invigilator's Signature: _____

Candidate's Signature: _____

Units covered: 1 Physical Quantities and Measurement • 2 Kinematics • 3 Dynamics-I • 4 Dynamics-II • 5 Pressure and Deformation • 6 Work and Energy (to Example 6.4) • 9 Nature of Science and Physics

— **END OF QUESTION PAPER** —